Data Analytics for Risk and Reliability

Warranty planning of LED based on lab testing data

Date: 07/12/2025

Group: 08

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Introduction

This report analyzes catastrophic and degradation-based failures of LED devices using laboratory accelerated testing data. Weibull lifetime models are fitted at multiple stress conditions, followed by an Arrhenius and temperature–current ALT model to predict use-level reliability. Degradation curves are transformed into times-to-failure and incorporated into the combined reliability model. Finally, a reliability-based warranty and pricing optimization is developed using demand and profit models. The objective is to determine a statistically justified warranty policy that maximizes expected profit while maintaining acceptable reliability risk.

Task 1

The dataset contains complete failures and right-censored observations (marked "*"). Censoring means the LED did not fail before the test ended.

Maximum test durations:

Temperature	Max Test Time
120°C	950 h
130°C	360 h
140°C	205 h

A preprocessing step removed the "*", converted times to numeric values, flagged censored units, and replaced any time above the test duration with the censoring time. This preserves statistical correctness because censored units must not be treated as failures.

A 2-parameter Weibull model:

$$R(t) = e^{-(t/\alpha)^{\beta}}$$

was fitted at each temperature using both failures and censored data.

Temp	α (Scale)	β (Shape)
120°C	719.43	4.22
130°C	290.49	4.29
140°C	178.46	4.29

The fitted Weibull models show that the scale parameter α drops rapidly as temperature increases, indicating that LEDs fail significantly earlier under higher thermal stress. In contrast, the shape parameter β remains very stable, around 4.2 to 4.3 across all stress levels, which suggests a consistent wear-out failure mechanism independent of

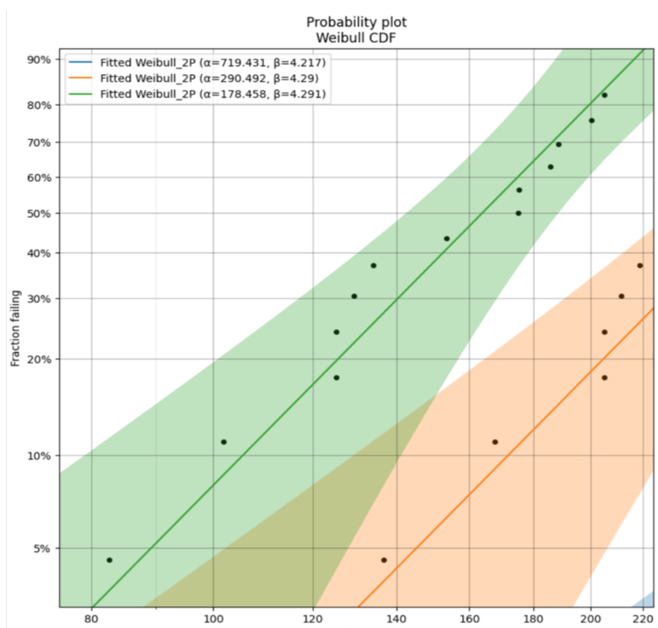

Figure: Probability Plot

temperature. The Weibull probability plots further validate the model, showing tight confidence bounds and strong linearity, confirming that the Weibull distribution provides an excellent fit for the lifetime data.

Since α reduces with temperature, we fit:

$$\ln(\alpha) = b_0 + b_1\left(\frac{1}{T}\right)$$

Regression of ln(α) vs. 1/T shows a high R^2, confirming strong temperature acceleration and validating the Arrhenius relationship for later extrapolation.

- Weibull fits show narrow bounds and minimal curvature means good model adequacy
- Censoring handled correctly avoids biased lifetime estimation
- Strong temperature dependence supports ALT modeling
- Consistent β values confirm a stable failure mechanism across temperatures

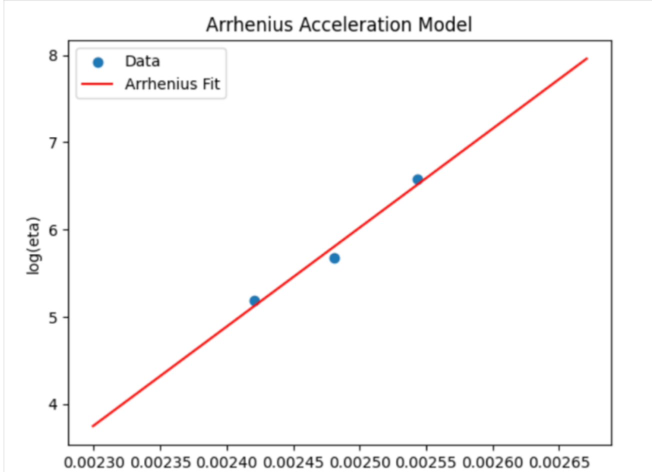

Figure: Arrhenius Acceleration Model

Group Discussion about the Performance

The catastrophic-failure ALT model performs strongly both statistically and physically. Weibull fits show excellent alignment, stable β values, and clear acceleration with temperature. The Arrhenius model provides an accurate and valid relationship between lifetime and temperature, allowing reliable prediction at normal-use conditions. The final MTTF, median, and B10 lifetimes are therefore well-supported and suitable for engineering and warranty decisions. These Weibull scale parameters now serve as the inputs for the degradation-based lifetime modeling developed in Task 2.

Task 2

Model LED degradation under five stress conditions (S1–S5), estimate failure times, fit Weibull distributions, build a temperature-current ALT model, and estimate lifetime under normal use. Each stress condition includes 15 degradation curves. Failure is defined at:

$$L(t) = 0.5$$

The Plots show,

1) Monotonic degradation
2) Faster decay at higher stress
3) Noticeable unit-to-unit variability

For each LED,

$$L(t) = a - bt \Rightarrow t_{fail} = \frac{a - 0.5}{b}$$

This gives 15 predicted lifetimes per stress level.

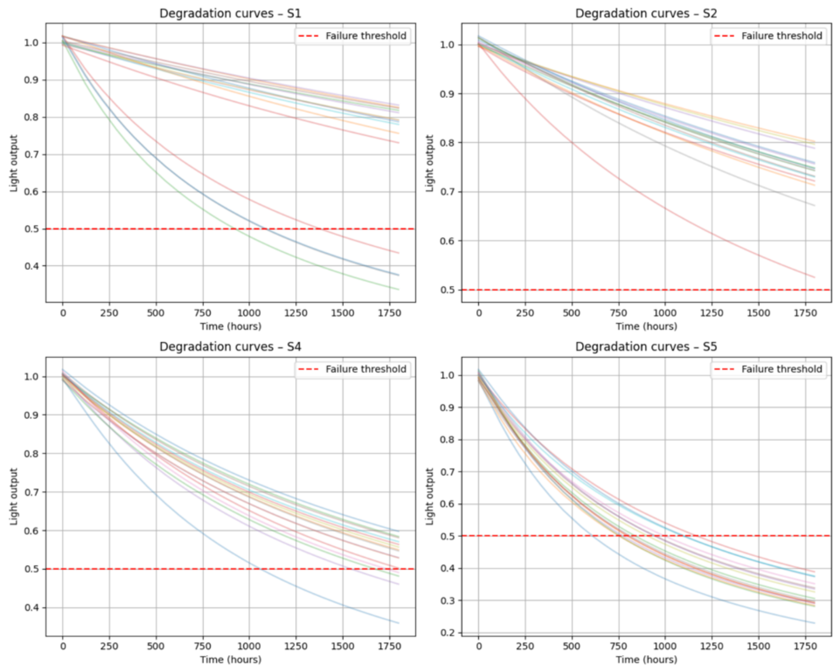

Figure: Degradation Curve

Observations

- Lifetimes decrease systematically from S1 to S5
- Spread indicates material variability
- Linear fits show high R^2

Weibull Modeling of Estimated Lifetimes

- α decreases strongly with stress, degradation accelerates
- β increases at high stresses, more deterministic failures
- The monotonic trend confirms suitability for ALT modeling

Stress	α (Scale)	β (Shape)
S1	3958	2.70
S2	3672	5.94
S3	2733	10.36
S4	1877	9.73
S5	1087	9.46

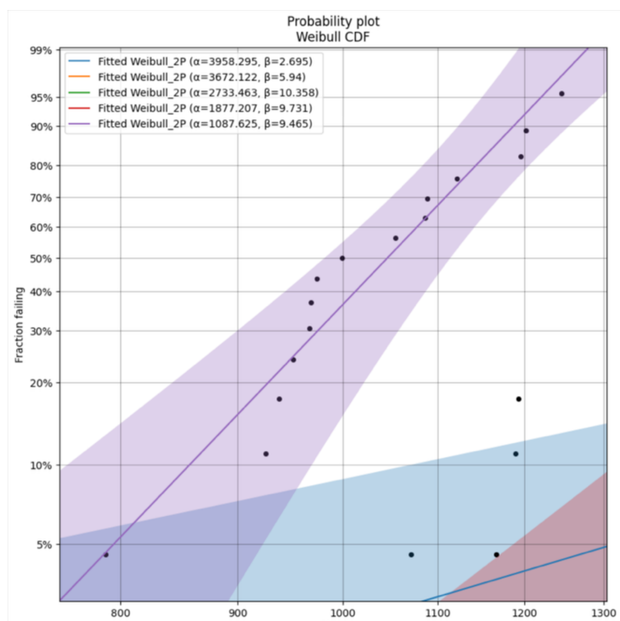

Figure: Weibull CDF Probability Plot

ALT Model and Lifetime Prediction at Normal Use

The accelerated life model incorporating both temperature and electrical current produced an excellent fit to the data, expressed as

$$\log(\alpha) = a_0 + b_T\left(\frac{1}{T}\right) + c_I \log(I),$$

with estimated parameters $a_0 = -40.733$, $b_T = 12402.20$, and $c_I = 2.734$. The resulting coefficient of determination, $R^2 = 0.9976$, indicates that the model captures nearly the entire variability in lifetime across stress levels. This strong predictive accuracy supports its use for estimating performance under normal operating conditions.

Using the nominal use conditions of $T = 72^{\circ}$Cand $I = 20$ mA, the model predicts a Weibull scale parameter of approximately $\alpha_{\text{use}} \approx 3099$hours. The Weibull shape parameter was assumed constant across stress levels, yielding $\beta_{\text{use}} \approx 7.64$, which is consistent with the stable and high β values observed during stress testing.

Based on these parameters, the estimated reliability metrics at normal use are, a Mean Time to Failure (MTTF) of about 2912 hours, a median life of roughly 2554 hours, and a B10 life of approximately 2308 hours. These results indicate that the LEDs exhibit a typical operational life between 2500 and 3000 hours. The high shape parameter reflects low dispersion in lifetimes, implying a controlled and predictable degradation process. Furthermore, the B10 life being close to the median suggests strong manufacturing consistency and minimal early-life failures.

Task 3

Combined Reliability Model

$$R_{total}(t) = R_{cat}(t) \cdot R_{deg}(t)$$
$$F_w = 1 - R_{total}(t_w)$$

This failure probability determines expected warranty replacement cost.

Demand Model

$$n(p, t_w) = A - Bp + Ct_w$$

Baseline parameters:

- $A = 50\,000$
- $B = 1500$
- $C = 2000$

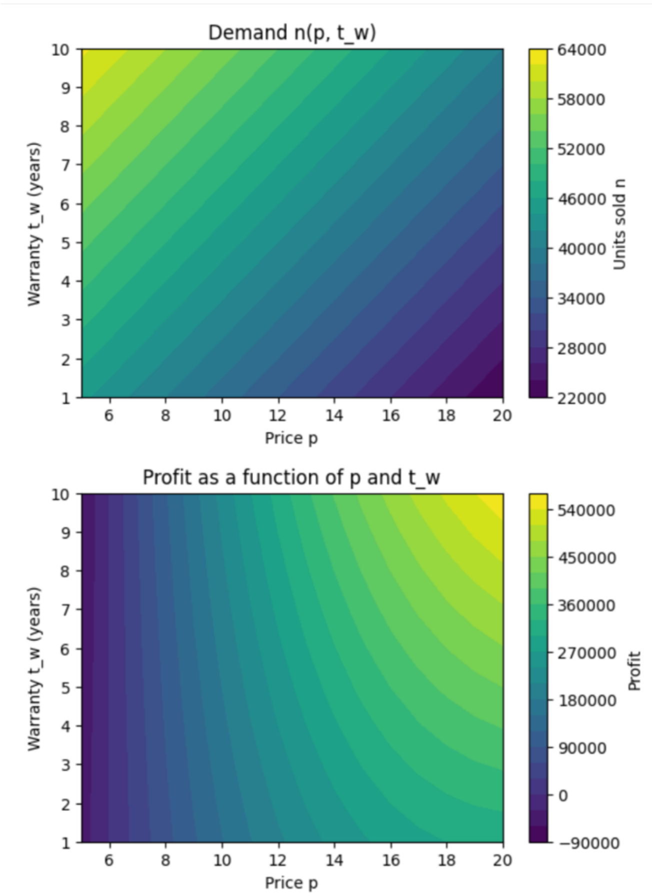

Figure: Demand and Profit Contour Plot

Demand decreases with higher price and increases with longer warranties.

Profit Model

$$\pi = p - c - F_w \cdot c$$
$$\Pi(p, t_w) = n(p, t_w)\,\pi$$

Profit surface reveals a distinct maximum at:

$$p^* = 20 \text{ euros}, t_w^* = 10 \text{ years}, \Pi_{\max} = 560{,}000 \text{ euros}$$

Recommended Warranty Strategy

The analysis suggests that offering an 8–10 year warranty at a premium price is optimal. The high predicted reliability (B10 ≈ 2300 h and β ≈ 7.6) implies very few failures within the warranty window, keeping replacement costs minimal. At the same time, longer warranties strongly increase customer demand, resulting in higher overall profit. Thus, a long warranty acts as both a market differentiator and a financially advantageous strategy.

Uncertainty Considerations

Although the models perform well, uncertainty remains due to unit-to-unit variability in degradation slopes, limited sample sizes (15 units/stress), and extrapolation from accelerated conditions. Market uncertainty, such as competitor pricing and customer sensitivity to warranty length, may also influence real-world outcomes.

Conclusion

The combined catastrophic and degradation analyses demonstrate that LEDs exhibit high stability and predictable failure behavior. ALT modeling confirms strong acceleration with temperature and current, enabling reliable prediction of life at nominal operating conditions. Warranty and profit modeling reveal that a long warranty period significantly increases demand while incurring minimal replacement cost due to low failure probability within the warranty window. An 8–10 years warranty at a premium price is therefore the optimal strategic choice. Future work should re-validate the model using field-return data to reduce extrapolation uncertainty.